

In the last two decades, there has been a complete transformation in every aspect of the user interface and the software value chain. Open source has established dominant players in various segments. For operating systems, we have Linux; in programming languages, PHP and Python stand out. MySQL leads in databases, Apache in servers, Firefox in browsers, and LibreOffice in office suites. The Android operating system, which needs no introduction, is everywhere. The community aspect has also flourished, with platforms like GitHub fostering numerous open source communities.

The journey of open source in India

Open source has impacted everything and is providing solutions daily. Whether you know it or not, open source is a part of everyone's life. Everyone using a mobile phone is consuming some form of open source. Here is how India is making a mark in this domain.

Open education: Education has benefited a lot from open source. MIT's OpenCourseWare is a pioneering example of open source educational content. With over a billion views, it has made high-quality educational material accessible to a global audience.

Similarly, in India, the NPTEL (National Programme on Technology Enhanced Learning) is an excellent example of the impact of open source on education. Coordinated by the IITs and IISc, NPTEL offers over 600 certification courses across 22 disciplines. It has become one of the largest educational programmes to be downloaded on YouTube, offering 56,000 hours of content in English and an additional 12,000 hours translated into various Indian languages.

These platforms exemplify how open source can democratise education, making high-quality

learning resources available to anyone with internet access.

Drug discovery: The Open Source Drug Discovery (OSDD) project, which began in Bengaluru, is an initiative funded by the Council of Scientific and Industrial Research (CSIR). CSIR oversees a network of 43 labs under its banner. Since its inception in 2008, OSDD has received about 35 million dollars in funding and has engaged nearly 500 contributors.

Drug discovery traditionally involved significant time and financial investments — typically around a billion dollars and seven years per drug. However, the OSDD project has significantly revolutionised this process. By leveraging the collective intelligence and collaborative efforts of a wide range of contributors from diverse fields, OSDD has managed to reduce both the cost and the time required for drug discovery. What used to take years now takes just a few months, with costs reduced drastically. This dramatic reduction in time and cost is a testament to the power of open source methodologies in pharmaceutical research.

ONDC: The core objective of the Open Network for Digital Commerce (ONDC) is to empower around 12 million small shopkeepers and sellers to harness the benefits of online presence. The aim is to simplify the process of going digital for these businesses. Instead of requiring complex software setups or the need for technical personnel, these shopkeepers can manage their online services by just pressing a few keys on their mobile phones. This level of accessibility and ease of use is what makes ONDC a powerful tool — and it's built on an innovative protocol, which is worth exploring.

Maya OS: The indigenous Maya OS operating system is expected to be used by all branches of the Indian Armed Forces. The adoption of Maya OS by such a critical and extensive

organisation is a significant step. It showcases the growing reliance on advanced open source technology and software within essential national sectors like defence, and underscores the importance of technological self-sufficiency and security in these areas.

Aadhaar- the national ID:

Aadhaar, the systems for which were initiated in India around 2009, was successfully deployed as a UID in 2014 and is a landmark project. Developed for the Indian government with taxpayers' money, it has achieved population-scale digital infrastructure that has been managed from within India.

Between 2009 and 2014, cloud technology matured significantly. While Aadhaar benefited from this, it couldn't fully utilise the cloud's capabilities then. Similarly, software development methodologies and microservices architectures were not as advanced as today, limiting Aadhaar's full potential in these aspects.

Beyond software: Open source has expanded beyond software into hardware. IIITB and NIMHANS are collaboratively contributing in this regard. Prof. Madhav Rao from IIITB and Dr Vikas Vizhayil, a neurosurgeon at NIMHANS, are together working on robotic brain surgery. An exoskeleton machine for individuals with severe back issues, who can't get up on their own, is being developed. It's designed to learn from the user's movements and walk with them. The machine is currently being tested, and the design will be open sourced, thanks to Prof. Rao's efforts.

Since 2018, our focus has also expanded to include chip-level design, with some of these designs being made open source. All this reflects a shift in thinking about open source — it's not just about software anymore. The artificial division between hardware and software is becoming obsolete.